**Remark 3.5** (Doubly stochastic kernel)**.** *Considering the Transformer dynamics on $\mathbb{R}^d$, thus without layer normalization, the authors in* [SABP22] *propose an alternative symmetric model: they replace the self-attention (stochastic) matrix by a doubly stochastic one, generated from the Sinkhorn iteration. This leads to a Wasserstein gradient flow, whereby the resulting attention mechanism is implicitly expressed as a limit of Sinkhorn iterations. Understanding the emergence of clusters for this model is an interesting but possibly challenging question.*

In view of the above discussion, we are inclined to propose the surrogate model

$$\dot{x}_i(t) = \mathbf{P}^{\perp}_{x_i(t)} \left( \frac{1}{n} \sum_{j=1}^{n} e^{\beta \langle x_i(t), x_j(t) \rangle} x_j(t) \right), \tag{USA}$$

which is obtained by replacing the partition function $Z_{\beta,i}(t)$ by $n$. As a matter of fact, (USA) presents a remarkably similar qualitative behavior—all of the results we show in this paper are essentially the same for both dynamics—.

The continuity equation corresponding to (USA), namely

$$\begin{cases} \partial_t \mu(t,x) + \operatorname{div} \left( \mathbf{P}^{\perp}_x \left( \int e^{\beta \langle x, x' \rangle} x' \, \mathrm{d}\mu(t,x') \right) \mu(t,x) \right) = 0 \\ \mu_{|t=0} = \mu_0 \end{cases} \tag{3.8}$$

for $(t,x) \in \mathbb{R}_{\geqslant 0} \times \mathbb{S}^{d-1}$, can now be seen as a Wasserstein gradient flow for the interaction energy $\mathsf{E}_\beta$ defined in (3.5).

**Lemma 3.6.** *Consider the interaction energy $\mathsf{E}_\beta : \mathcal{P}(\mathbb{S}^{d-1}) \to \mathbb{R}_{\geqslant 0}$ defined in* (3.5)*. Then the vector field*

$$\mathcal{X}[\mu](x) = \mathbf{P}^{\perp}_x \left( \int e^{\beta \langle x, x' \rangle} x' \, \mathrm{d}\mu(x') \right)$$

*satisfies*

$$\mathcal{X}[\mu](x) = \nabla \delta \mathsf{E}_\beta[\mu](x) \tag{3.9}$$

*for any $\mu \in \mathcal{P}(\mathbb{S}^{d-1})$ and $x \in \mathbb{S}^{d-1}$, where $\delta \mathsf{E}_\beta[\mu]$ denotes the first variation of $\mathsf{E}_\beta$.*

We omit the proof which follows from standard Otto calculus [Ott01], [Vil09, Chapter 15], [CNWR24, Chapter 5]. We can actually write (3.9) more succinctly by recalling the definition of the convolution of two functions on $\mathbb{S}^{d-1}$ [DX13, Chapter 2]: for any $g \in L^1(\mathbb{S}^{d-1})$ and $f : [-1,1] \to \mathbb{R}$ such that $t \mapsto (1-t^2)^{\frac{d-3}{2}} f(t)$ is integrable,

$$(f * g)(x) = \int f(\langle x, y \rangle) g(y) \, \mathrm{d}\sigma_d(y).$$

This definition has a natural extension to the convolution of a function $f$ (with the above integrability) and a measure $\mu \in \mathcal{P}(\mathbb{S}^{d-1})$. We can hence rewrite

$$\mathsf{E}_\beta[\mu] = \frac{1}{2} \int (\mathsf{G}_\beta * \mu)(x) \, \mathrm{d}\mu(x)$$

where $[-1,1] \ni \mathsf{G}_\beta(t) = \beta^{-1} e^{\beta t}$, and so

$$\mathcal{X}[\mu](x) = \nabla (\mathsf{G}_\beta * \mu)(x).$$

Thus, (3.8) takes the equivalent form

$$\begin{cases} \partial_t \mu(t,x) + \operatorname{div}\Big(\nabla\big(\mathsf{G}_\beta * \mu(t,\cdot)\big)(x)\mu(t,x)\Big) = 0 & \text{for } (t,x) \in \mathbb{R}_{\geqslant 0} \times \mathbb{S}^{d-1} \\ \mu_{|t=0} = \mu_0 & \text{for } x \in \mathbb{S}^{d-1}. \end{cases} \tag{3.10}$$

The considerations above lead us to the following Lyapunov identity.

**Lemma 3.7.** *The solution $\mu \in C^0(\mathbb{R}_{\geqslant 0}; \mathcal{P}(\mathbb{S}^{d-1}))$ to (3.8) satisfies*

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathsf{E}_\beta[\mu(t)] = \int \Big\|\nabla\Big(\mathsf{G}_\beta * \mu(t,\cdot)\Big)(x)\Big\|^2 \,\mathrm{d}\mu(t,x)$$

*for $t \geqslant 0$.*

Interestingly, (3.10) is an *aggregation* equation, versions of which have been studied in great depth in the literature. For instance, clustering in the spirit of an asymptotic collapse to a single Dirac measure located at the center of mass of the initial density $\mu(0,\cdot)$ has been shown for aggregation equations with singular kernels in [BCM08, BLR11, CDF+11], motivated by the Patlak-Keller-Segel model of chemotaxis. Here, one caveat (and subsequently, novelty) is that (3.10) is set on $\mathbb{S}^{d-1}$ which makes the analysis developed in these references difficult to adapt or replicate.

**Remark 3.8** (Particle version)**.** *Let us briefly sketch the particle version of the Wasserstein gradient flow (3.8). When $\mu(t) = \frac{1}{n}\sum_{i=1}^n \delta_{x_i(t)}$, the interaction energy (3.5) takes the form*

$$\mathsf{E}_\beta(X) = \frac{1}{2\beta n^2}\sum_{i=1}^n\sum_{j=1}^n e^{\beta\langle x_i, x_j\rangle}$$

*where $X = (x_1,\dots,x_n) \in (\mathbb{S}^{d-1})^n$. Denoting by $\nabla_X$ the gradient associated to the standard Riemannian metric on $(\mathbb{S}^{d-1})^n$, we get the dynamics*

$$\dot{X}(t) = n\nabla_X \mathsf{E}_\beta(X(t)). \tag{3.11}$$

*Indeed, the gradient on $(\mathbb{S}^{d-1})^n$ is simply $\nabla = (\partial_1,\dots,\partial_n)$ where $\partial_i$ is the gradient in $\mathbb{S}^{d-1}$ acting on the $i$-th copy in $(\mathbb{S}^{d-1})^n$. Therefore*

$$\partial_i \mathsf{E}_\beta(X(t)) = \frac{1}{\beta n^2}\sum_{j=1}^n \mathsf{P}^\perp_{x_i(t)}\left(e^{\beta\langle x_i(t), x_j(t)\rangle}\beta x_j(t)\right) = \frac{1}{n}\dot{x}_i(t)$$

*which yields (3.11).*

*Note that (SA) also corresponds to a gradient flow of the same interaction energy albeit with respect to a Riemannian metric on the sphere different from the standard one (for $n = 2$ the two are conformally equivalent). We provide more detail in the following section.*

3.4. (SA) **is a gradient flow for a modified metric.** We will now briefly demonstrate that for a particular choice of parameters $(Q, K, V)$, the true dynamics (SA) can be seen as a gradient flow for $\mathsf{E}_\beta$ upon a modification of the metric on the tangent space of $(\mathbb{S}^{d-1})^n$. This will facilitate qualitative analysis later on by using standard tools from dynamical systems. This insight can then be extrapolated to the corresponding continuity equation (3.4) as well, as seen in Section 3.4.2.

3.4.1. *The case of particles.* We suppose that

$$Q^\top K \text{ is symmetric}, \qquad V = Q^\top K.$$

We define a new metric on $(\mathbb{S}^{d-1})^n$ as follows. Let $X = (x_1, \dots, x_n) \in (\mathbb{S}^{d-1})^n$. Consider the inner product on $\mathrm{T}_X(\mathbb{S}^{d-1})^n$ given by

$$\langle (a_1, \dots, a_n), (b_1, \dots, b_n) \rangle_X = \sum_{i=1}^n Z_{\beta,i}(X) \langle a_i, b_i \rangle, \tag{3.12}$$

where $a_i, b_i \in \mathrm{T}_{x_i}\mathbb{S}^{d-1}$, and

$$Z_{\beta,i}(X) = \sum_{j=1}^n e^{\beta \langle V x_i, x_j \rangle}.$$

Set

$$\mathsf{E}_\beta(X) = \frac{1}{2\beta} \sum_{i=1}^n \sum_{j=1}^n e^{\beta \langle V x_i, x_j \rangle}.$$

We now show that the dynamics (2.3) can be equivalently written as

$$\dot{X}(t) = \nabla \mathsf{E}_\beta(X(t)),$$

where the gradient $\nabla$ is computed with respect to the metric (3.12) on $(\mathbb{S}^{d-1})^n$. To this end, we ought to show that for all vector fields $Y$ on $(\mathbb{S}^{d-1})^n$ and for all $X \in (\mathbb{S}^{d-1})^n$,

$$\frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=0} \mathsf{E}_\beta\left(\Phi_Y^t(X)\right) = \langle Y(X), B(X) \rangle_X \tag{3.13}$$

holds, where $\Phi_Y^t$ is the flow associated to the vector field $Y$, whereas $B = (B_1, \dots, B_n)$ with

$$B_i = \mathbf{P}_{x_i}^\perp \left( \frac{1}{Z_{\beta,i}(X)} \sum_{j=1}^n e^{\beta \langle V x_i, x_j \rangle} V x_j \right) \in \mathrm{T}_{x_i}\mathbb{S}^{d-1}.$$

By linearity, it is sufficient to show (3.13) for vector fields $Y$ of the form

$$Y(X) = (Ax_1, 0, \dots, 0) \in \mathrm{T}_X(\mathbb{S}^{d-1})^n$$

where $A$ is an arbitrary non-zero skew-symmetric matrix. Clearly

$$\Phi_Y^t(X) = (e^{tA} x_1, x_2, \dots, x_n). \tag{3.14}$$

One first computes

$$\frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=0} \mathsf{E}_\beta\left(\Phi_Y^t(X)\right) = \sum_{j=1}^n e^{\beta \langle V x_1, x_j \rangle} \langle A x_1, V x_j \rangle.$$

Now observe that $\langle A x_1, y \rangle = \langle A x_1, z \rangle$ for all skew-symmetric matrices $A$ if and only if $x_1(y^\top - z^\top)$ is a symmetric matrix. Since $\mathbf{P}_{x_1}^\perp = I_d - x_1 x_1^\top$, we see that

$$\sum_{j=1}^n e^{\beta \langle V x_1, x_j \rangle} \langle A x_1, V x_j \rangle = \langle Y(X), B(X) \rangle_X,$$

as desired.